ELEC60022
EE9-AO12

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE PART III/IV: MEng, BEng.and ACGI

MICROWAVE TECHNOLOGY

Wednesday, 11 December 2024, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer question ONE. Answer TWO out of the remaining three questions.

Question ONE carries a weighting of 50%. The rest carry a weighting of 25% each.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : K. Fobelets

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Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

There are 4 (four) questions in total.

Question 1 is COMPULSORY and carries a weighting of 50%.

Answer 2 (two) out of the remaining 3(three) questions, each carrying a weighting of 25%.

All variables and abbreviations have their usual meaning.

1.

a) Sketch a symmetrical stripline, showing all relevant variables. Define each variable and give their units.

[5]

b) Compare and contrast stripline with microstrip, in terms of electromagnetic characteristics and practical implementations.

[5]

c) Assume any ground plane to be a perfect electrical conductor. The signal track has a width of 100 μm (equal to the total dielectric thickness divided by 2π) and its $skin\ depth \ll metal\ thickness \ll total\ dielectric\ thickness$. The signal track is aluminium (having a DC conductivity of $\sigma_o = 3.774 \times 10^7$ S/m) and the dielectric is polyimide (having an effective relative permittivity of ε_r = 2.8 – j 0.039). Unless otherwise stated, calculate the following at 10 GHz and include all units; clearly stating any simplifying assumptions:

i) Total dielectric thickness and effective signal track width. [3]

ii) Lossless characteristic impedance. [2]

iii) Lossless phase velocity. [2]

iv) Lossless phase constant. [2]

v) Distributed capacitance. [3]

vi) Distributed conductance. [3]

vii) Distributed inductance. [2]

viii) Skin depth in the signal track. [2]

ix) Distributed resistance. [2]

x) Lossy characteristic impedance. [3]

xi) Lossy propagation constant. [3]

xii) Attenuation constant due to the conductors only. [2]

xiii) Attenuation constant due to the dielectric only. [2]

xiv) Total attenuation constant and insertion loss in dB/m. [3]

xv) Q-factor associated with the conductors only. [2]

xvi) Q-factor associated with the dielectric only. [2]

xvii) Overall Q-factor. [2]

2.

Consider two air-filled metal-pipe rectangular waveguide (MPRWG) standards, having the broad wall dimension a and narrow wall dimension b:

WR10: a = 2.540 mm and b = 1.270 mm
WR8: a = 2.032 mm and b = 1.016 mm

i) Sketch the typical S-parameter frequency characteristics for a typical low-loss MPRWG, clearly indicating the regions below and above its cut-off frequency.

[4]

ii) Calculate the lowest cut-off frequencies for both the WR10 and WR8 MPRWGs.

[4]

iii) Calculate the guided-wavelength for the WR8 waveguide at 100 GHz.

[4]

iv) If a WR8 MPRWG is connected between two WR10 MPRWGs, how long would the RW8 MPRWG need to be to excite the dominant Fabry-Perot resonance at 100 GHz?

[2]

v) For the Fabry-Perot resonator, described in 2(iv), what is the coupling mechanism?

[3]

vi) Calculate the wave impedances for both the WR10 and WR8 MPRWGs at 100 GHz.

[4]

vii) If a WR8 MPRWG is connected to a reference WR10 MPRWG feed line, at 100 GHz, calculate the voltage-wave reflection coefficient and return loss at the connecting flanges.

[4]

3.

Consider the DC biasing of a microwave transistor. An ideal voltage-source power supply is connected to a shunt metal-insulator-metal decoupling capacitor, via a series choke inductor. Both the choke inductor and the decoupling capacitor are made from PEC. An ideal connection is made between the decoupling capacitor and the microwave transistor. Clearly state any assumptions made.

a) The choke inductor is made using a straight piece of 1 cm long and 1 mm diameter wire, suspended in air, having an inductance L_o. This wire is then encased within a zinc–nickel ferrite, having relative permeability $\mu_r = 3.1 - j\ 1.9$:

   i) Assuming this permeability is frequency independent, sketch the equivalent circuit model for this choke inductor.
   ii) Calculate all corresponding values at 1 GHz.
   iii) Calculate the associated unloaded Q-factor Q_m from this circuit and compare this with that from the ferrite material.

[6]

b) The decoupling capacitor is formed using a 1 cm × 1 cm top plate and the bottom plate employs the ground plane on the other side of a 635 μm thick alumina substrate, having an effective relative permittivity of $\varepsilon_{reff} = 9.8 - j\ 0.0001$:

   i) Assuming this permeability is frequency independent, sketch the equivalent circuit model for this decoupling capacitor.
   ii) Calculate all corresponding values at 1 GHz.
   iii) Calculate the associated unloaded Q-factor Q_m from this circuit and compare this with that from the substrate material.

[6]

c) By combining the equivalent circuit models for the power supply, choke inductor and decoupling capacitor, sketch the equivalent circuit model of the biasing network as an equivalent shunt *RLC*-network and calculate the driven resonance frequency. *Hint: first transform the series choke inductor model into its parallel equivalent circuit model and recalculate the associated resistances at the driven resonance frequency.*

[6]

d) Using the *RLC* element values determined in 3(b) and 3(c):

   i) Calculate the overall unloaded Q-factor Q_o of the network and compare this to that using the values of Q_m and Q_e (determined previously).
   ii) Calculate the complex resonance frequency and from this calculate the unloaded Q-factor.

[4]

e) Using the complex resonance frequency determined in 3(c), at the driven resonance frequency, calculate the return loss for the DC biasing network 'seen' by the microwave transistor having an impedance of 66.7 Ω. What can be inferred, in terms of the useful bandwidth of the circuit?

[3]

4.

This question relates to directional couplers.

a) For a 3 dB waveguide coupler:

i) Sketch the coupler in rectangular waveguide, clearly indicating the port numbers and give the usual name attributed to each port. Briefly describe how it works.
[3]

ii) By inspection, or otherwise, write down the associated lossless S-parameter matrices for the centre frequency of operation.
[1]

iii) What is the requirement for the port termination impedances?
[2]

iv) State at least two applications for this coupler.
[2]

b) For a 3 dB coupled-line coupler:

i) Sketch the coupler in microstrip, clearly indicating the port numbers and give the usual name attributed to each port.
[2]

ii) By inspection, or otherwise, write down the associated lossless S-parameter matrices for the centre frequency of operation.
[1]

iii) What is the requirement for the port termination impedances?
[2]

iv) State at least two applications for the this coupler.
[2]

c) For a 3 dB Marchand Balun:

i) Sketch its symbolic representation, clearly indicating the port numbers and give the usual name attributed to each port.
[1]

ii) By inspection, or otherwise, write down the associated lossless S-parameter matrices for both the centre frequency of operation and at DC.
[1]

iii) What is the requirement for the port termination impedances?
[1]

iv) Using coupled-line couplers, show how the balun can be implemented by sketching its complete circuit.
[3]

v) How can the coupler in 4(b) be physically modified in order to be used in the balun?
[2]

vi) State at least two applications for this balun.
[2]